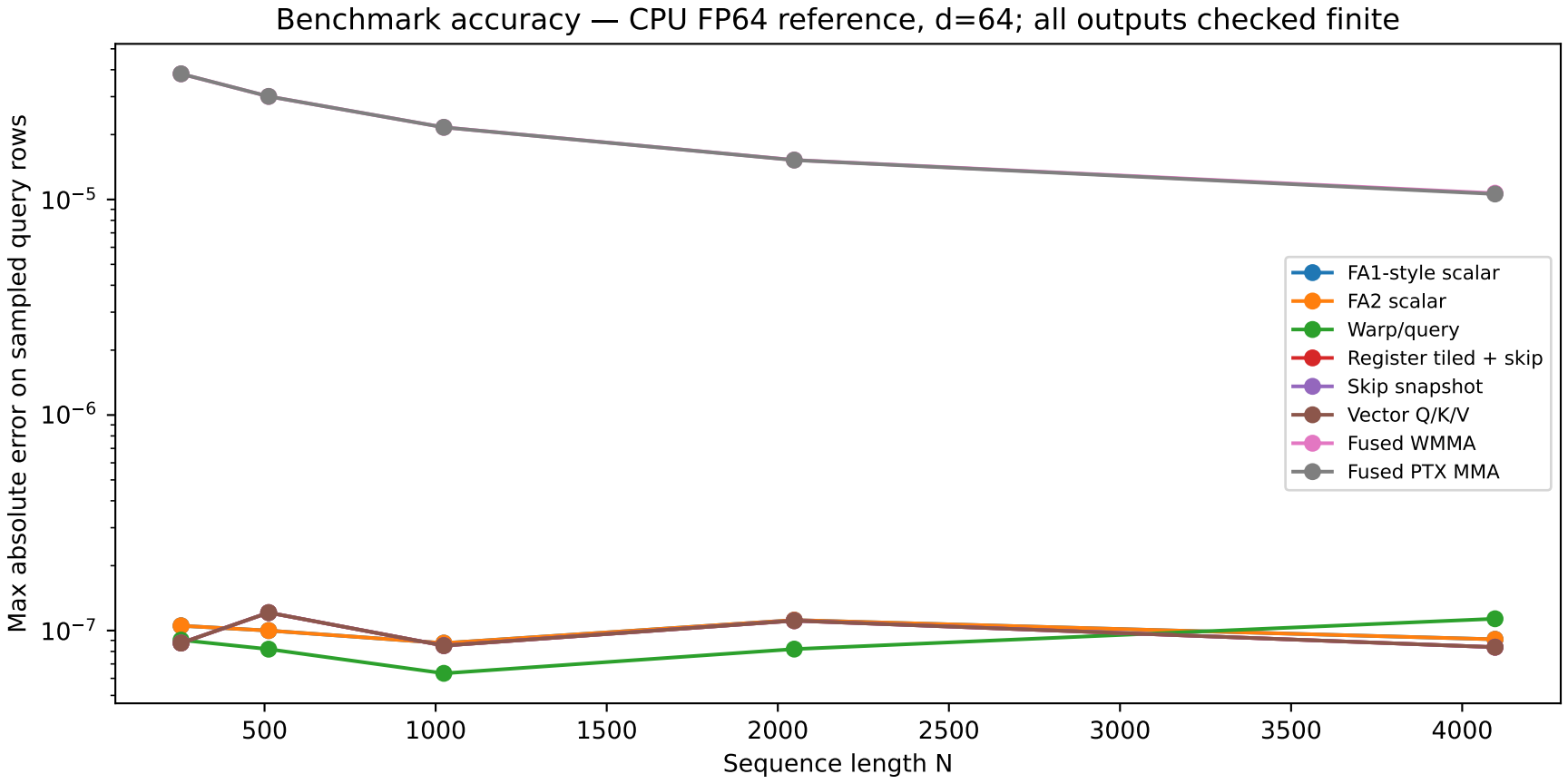
Benchmark accuracy — CPU FP64 reference, d=64; all outputs checked finite
Max absolute error on sampled query rows
$10^{-5}$
$10^{-6}$
$10^{-7}$
500
1000
1500
2000
2500
3000
3500
4000
Sequence length N
FA1-style scalar
FA2 scalar
Warp/query
Register tiled + skip
Skip snapshot
Vector Q/K/V
Fused WMMA
Fused PTX MMA